



Std NAME: 30/40

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| a | b | b | b | b | b | a | c | d | d  | d  | d  | d  | d  | d  | a  | c  | d  | c  | d  | c  | b  | d  | d  | d  |

(12 points)

Q1. Choose the best answer (fill your answer in the table)

- A list of protocols used by a system, one protocol per layer, is called  
a) protocol architecture b) protocol stack c) protocol suite d) none of the mentioned
- In TCP IP Model, when data is sent from device A to device B, the 5th layer to receive data at B is  
a) Application layer b) Transport layer c) Link layer d) Session layer
- In the OSI model, as a data packet moves from the lower to the upper layers, headers are \_\_\_\_\_  
a) Added b) Removed c) Rearranged d) Discarded
- Which layer is responsible for process to process delivery?  
a) network layer b) transport layer c) session layer d) data link layer
- Transmission data rate is decided by  
a) network layer b) physical layer c) data link layer d) transport layer
- Transport layer aggregates data from different applications into a single stream before passing it to  
a) network layer b) data link layer c) application layer d) physical layer
- The time required to examine the packet's header and determine where to direct the packet is part of  
a) Processing delay b) Queuing delay c) Transmission delay d) All of the mentioned
- A local telephone network is an example of a \_\_\_\_\_ network.  
a) Packet switched b) Circuit switched c) Both Packet switched and Circuit switched d) Line switched
- Most packet switches use this principle \_\_\_\_\_  
a) Stop and wait b) Store and forward c) Both Stop and wait and Store and forward d) Stop and forward
- What are the Methods to move data through a network of links and switches?  
a) Packet switching b) Circuit switching c) Both Packet switching and Circuit switching d) Line switching
- In \_\_\_\_\_ resources are allocated on demand.  
a) packet switching b) circuit switching c) line switching d) frequency switching
- Which one of the following is an architecture paradigms?  
a) Peer to peer b) Client-server c) HTTP d) Both Peer-to-Peer & Client-Server
- E-mail is \_\_\_\_\_  
a) Loss-tolerant application b) Bandwidth-sensitive application c) Elastic application d) None of the mentioned
- The \_\_\_\_\_ translates internet domain and host names to IP address.  
a) domain name system b) routing information protocol c) network time protocol d) internet relay chat
- Application layer protocol defines \_\_\_\_\_  
a) types of messages exchanged b) message format, syntax and semantics c) rules for when and how processes send and respond to messages d) all of the mentioned
- Which one of the following protocol delivers/stores mail to receiver server?  
a) simple mail transfer protocol b) post office protocol c) internet mail access protocol d) hypertext transfer protocol
- When displaying a web page, the application layer uses the \_\_\_\_\_  
a) HTTP protocol b) FTP protocol c) SMTP protocol d) TCP protocol

18. The default connection type used by HTTP is \_\_\_\_\_  
 a) Persistent  
c) Can be either persistent or non-persistent depending on connection request  
 d) None of the mentioned
19. The \_\_\_\_\_ method when used in the method field, leaves entity body empty.  
 a) POST  
 b) SEND  
 c) GET  
 d) PUT
20. Multiple object can be sent over a TCP connection between client and server in  
 a) persistent HTTP  
c) both persistent HTTP and nonpersistent HTTP  
 b) nonpersistent HTTP  
 d) none of the mentioned
21. DNS database contains \_\_\_\_\_  
 a) name server records  
 b) hostname-to-address records  
d) all of the mentioned  
 c) hostname aliases
22. To join the internet, the computer has to be connected to a  
 a) internet architecture board  
 b) internet society  
c) internet service provider  
 d) none of the mentioned
23. Internet access by transmitting digital data over the wires of a local telephone network is provided by  
 a) leased line  
 b) digital subscriber line  
 c) digital signal line  
d) none of the mentioned
24. MAC address is of  
a) 24 bits  
 b) 36 bits  
 c) 42 bits  
d) 48 bits

## Q2. Define the following terms

Computer network, Protocol, ISP, 3 way hand shake, connection less, packet, access network, core network, OSI, link, network device, stateless protocol, the internet, www, socket, TCP/IP,

## Q3. Detailed Answers

1. (1 pts) Why HTTP server is stateless NOT state-full? (0.5)
2. (1 pts) User-server state: How cookies are used to keeping state? (1)
3. (2 pts) What cookies can be used for? (2)
4. (2 pts) Web caches (proxy server): (1.5)
  - a. What is its goal? ✓
  - b. Explain how the proxy server acts as client and server at the same time? ✓
  - c. Where it is installed? ✓
  - d. Why web caching? ✓
5. (2pts) Application architecture are classified as client server and P2P. Explain them? (2)
6. (3 pts) Explain how DNS, HTTP, and SMTP works?. What applications they serve? (2)
7. (4 pts) Compare packet switching and Circuit switching? (4)

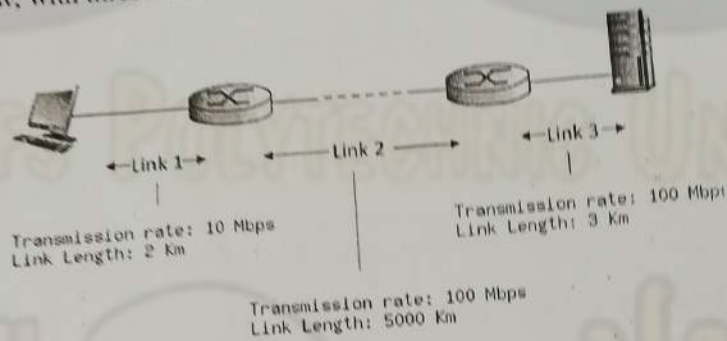
(15 points)

(5 points)

## Q4. Answer ANY two from the following

## 1. Computing end-end delay (transmission and propagation delay)

Consider the figure below, with three links, each with the specified transmission rate and link length.

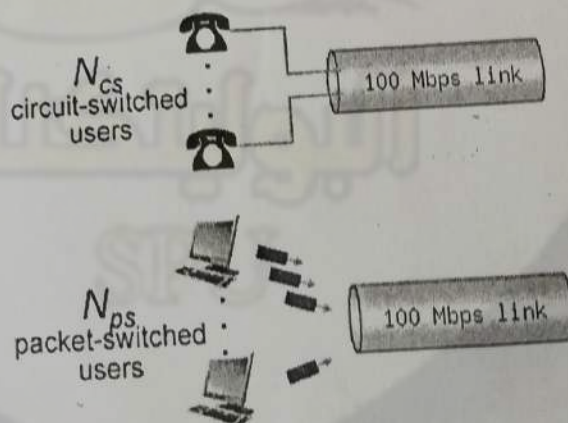


Find the end-to-end delay (including the transmission delays and propagation delays on each of the three links, but ignoring queueing delays and processing delays) from when the left host begins transmitting the first bit of a packet to the time when the last bit of that packet is received at the server at the right. The speed of light propagation delay on each link is  $3 \times 10^8$  m/sec. Note that the transmission rates are in Mbps and the link distances are in Km. Assume a packet length of 4000 bits. Give your answer in milliseconds.

## 2. Quantitative Comparison of Packet Switching and Circuit Switching

This question requires a little bit of background in probability (but we'll try to help you though it in the solutions). Consider the two scenarios below:

- A circuit-switching scenario in which  $N_{cs}$  users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 100 Mbps.
- A packet-switching scenario with  $N_{ps}$  users sharing a 100 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 10 percent of the time.



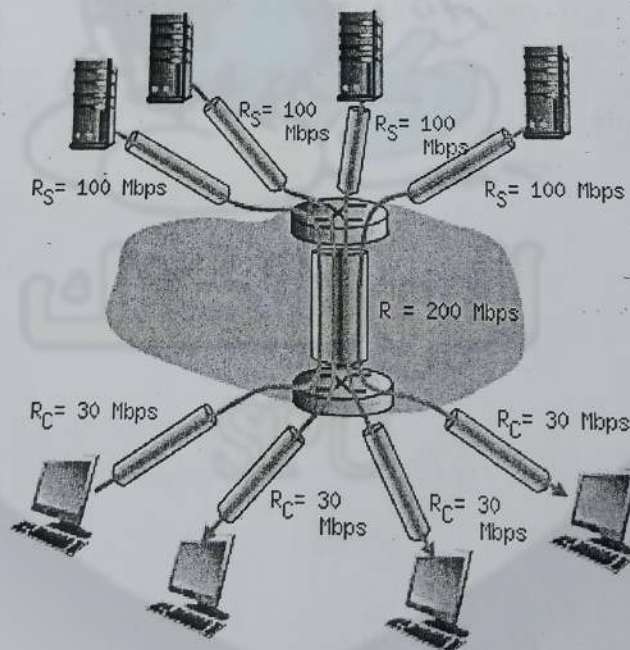
Answer the following questions:

- When circuit switching is used, what is the maximum number of circuit-switched users that can be supported? Explain your answer.
- For the remainder of this problem, suppose packet switching is used. Suppose there are 7 packet-switching users (i.e.,  $N_{ps} = 7$ ). Can this many users be supported under circuit-switching? Explain.
- What is the probability that a given (*specific*) user is transmitting, and the remaining users are not transmitting?
- What is the probability that one user (*any* one among the 7 users) is transmitting, and the remaining users are not transmitting? When one user is transmitting, what fraction of the link capacity will be used by this user?
- What is the probability that any 4 users (of the total 7 users) are transmitting and the remaining users are not transmitting? (Hint: you will need to use the binomial distribution [1, 2]).
- What is the probability that *more* than 4 users are transmitting? Comment on what this implies about the number of users supportable under circuit switching and packet switching.

### 3. End to End Throughput and Bottleneck Links

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of  $R = 200$  Mbps. The four links from the servers to the shared link have a transmission capacity of  $R_S = 100$  Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of  $R_C = 30$  Mbps per second. You might want to review Figure 1.20 in the text before answering the following questions:

- What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fair-shared (i.e., divides its transmission rate equally among the four pairs)?
- Which link is the bottleneck link for each session?
- Assuming that the senders are sending at the maximum rate possible, what are the link utilizations for the sender links ( $R_S$ ), client links ( $R_C$ ), and the middle link ( $R$ )?



Good Luck



**Question 1:**

1. There are the following statements that are given below, which of them are correct about the OSI model?
  - A. The OSI model is a reference model that describes the network functionalities.
  - B. The OSI model is an implemented model that describes only network protocols.
  - C. The OSI model is an implemented model that describes only network topologies.
  - D. None of the above.
2. Which layer of the OSI model is responsible for actual physical connections among the network devices?
  - A. Data Link Layer
  - B. Network Layer
  - C. Transport Layer
  - D. Physical Layer
3. Which of the following are functions of the application layer?
  - A. Mail Services
  - B. File Transfer
  - C. Virtual terminals
  - D. All of the above
4. Which layer of the OSI model is responsible for segmentation and reassembly?
  - A. Data Link Layer
  - B. Network Layer
  - C. Transport Layer
  - D. Physical Layer
5. There are the following statements that are given below, which of them are correct about HTTP?
  - A. HTTP is media independent protocol
  - B. It is a stateless protocol
  - C. It is a connectionless protocol
  - D. HTTP is mainly used to fetch HTML documents
  1. A, B, C, and D
  2. A, B, and C
  3. B, C, and D
  4. A, C, and D
6. What an HTTP message contains?
  - A. Message Header
  - B. Message Type
  - C. Message Body
  - D. Message Length
  1. A, C, and D
  2. A, B, and C
  3. B, C, and D
  4. A, B, C, and D
7. Which of the following are the types of HTTP messages?
  - A. HTTP header message
  - B. HTTP request message
  - C. HTTP response message
  - D. HTTP footer message
  1. A and D
  2. B and C
  3. Only B
  4. A, B, C, and D
8. There are the following statements that are given below, which of them are correct about HTTPS?
 

- A. HTTPS is the extended version of HTTP.
  - B. HTTPS is used for secured communication between the web browser and web server.
  - C. HTTPS stands for HyperText Transfer Protocol Secure.
  - D. HTTPS also provides anti-virus security for the website.
  1. A and B
  2. A and C
  3. A, B, and C
  4. A, B, C, and D
9. There are the following statements that are given below, which of them are correct about the advantages of HTTPS?
  - A. Data Integrity
  - B. Encryption based security
  - C. Search Engine Optimization.
  - D. Antivirus protection
  1. A and B
  2. A and C
  3. A, B, and C
  4. A, B, C, and D



10. What are the core components of the email system?

|                   |                   |                              |                  |
|-------------------|-------------------|------------------------------|------------------|
| A. Email client   | B. Email server   | C. Internet service provider | D. All the above |
| 1. <u>A and B</u> | 2. <u>B and D</u> | 3. <u>A and C</u>            | 4. <u>D</u>      |

11. There are following statements that are given below, which of them are correct about email addresses?

|                                         |                                                           |                                                                                             |                                           |
|-----------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------|
| A. E-mail addresses are case sensitive. | B. E-mail address contains the domain name and user name. | C. In the e-mail address, the user name and domain name are separated by the dollar symbol. | D. E-mail addresses cannot be duplicated. |
| 1. <u>A and B</u>                       | 2. <u>B and D</u>                                         | 3. <u>A, B and C</u>                                                                        | 4. <u>A, B, and D</u>                     |

12. Which of the following protocols are used for email?

|                   |                   |                      |                       |
|-------------------|-------------------|----------------------|-----------------------|
| A. SMTP           | B. POP            | C. ICMP              | D. IMAP               |
| 1. <u>A and B</u> | 2. <u>A and C</u> | 3. <u>A, B and C</u> | 4. <u>A, B, and D</u> |

13. There are following statements that are given below, which of them are correct about IMAP?

|                                                                                              |                                                          |                                             |                   |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------|-------------------|
| A. The IMAP is used to manipulate email messaging without downloading it on a local machine. | B. It is used to access multiple mailboxes concurrently. | C. IMAP enables users to search for emails. | D. All the above. |
| 1. <u>A and B</u>                                                                            | 2. <u>A and C</u>                                        | 3. <u>B and C</u>                           | 4. <u>D</u>       |

14. There are following statements that are given below, which of them are correct about POP?

|                                         |                                            |                                             |                     |
|-----------------------------------------|--------------------------------------------|---------------------------------------------|---------------------|
| A. POP stands for Post Office Protocol. | B. POP is used to support a single client. | C. POP supports offline access to messages. | D. All of the above |
| 1. <u>A and B</u>                       | 2. <u>A and C</u>                          | 3. <u>B and C</u>                           | 4. <u>D</u>         |

15. There are the following statements that are given below, which of them are correct about URL?

|                                               |                                                              |                                                                         |                                                         |
|-----------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------|
| A. URL stands for Universal Resource Locator. | B. URL is a global address to access resources over the Web. | C. The resources accessed by the URL can be an HTML file, CSS file, etc | D. URL cannot use any protocol to access web resources. |
| 1. <u>A and B</u>                             | 2. <u>B and C</u>                                            | 3. <u>A, B, and C</u>                                                   | 4. <u>A, B, C, and D</u>                                |

16. Which of the following schemes can be used in a URL?

|                   |                       |                   |                          |
|-------------------|-----------------------|-------------------|--------------------------|
| A. http://        | B. telnet://          | C. file://        | D. ftp://                |
| 1. <u>A and B</u> | 2. <u>A, B, and D</u> | 3. <u>A and D</u> | 4. <u>A, B, C, and D</u> |

- Answer:



**Question 2:**

1. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable? (2 marks)

2. What information is used by a process running on one host to identify a process running on another host? (2 marks)

3. What transport service does an application need? (2marks)

4. HTTP protocol is stateless. What means the stateless? (mark)

**Good luck**

**Question 1: True / False**

(3.5 marks)

1. There is no difference between a host and an end system. In the internet, all devices are called hosts and end systems. So, hosts and end systems are used interchangeably. ( )
2. A Web server is a program that uses HTTP (Hypertext Transfer Protocol) to serve the files that form Web pages to users, in response to their requests, which are forwarded by their computer's HTTP clients. Dedicated computers and appliances may be referred to as Web servers as well. ( )
3. Packet switching is a method of transferring the data to a network in form of packets. In order to transfer the file fast and efficiently manner over the network and minimize the transmission latency, the data is broken into small pieces of variable length. ( )
4. Store and forward switching technique, when the switch receives the frame, it stores the frame data in buffers until the full frame has been received. During the process, the switch analyses the current frame for information about its destination. ( )
5. Client-Server Architecture is an always on host called the server, which services requests from many other hosts, called clients. Client hosts can be sometimes on or always on. With the client-server architecture, clients do not directly communicate with each other. Client-server architecture's server has a fixed, well known address called an IP address. These characteristics allow a client the ability to always contact the server by sending a packet to the server's address. ( )
6. The goal of Proxy server is satisfy client request without involving original server. ( )
7. The Domain Network System (DNS) protocol helps Internet users and network devices discover websites using human-readable hostnames, instead of numeric IP addresses. ( )

**Question 2:**

1. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable? (4 marks)

What is the difference between network architecture and application architecture?

(2)



**Question 3:**

1. What information is used by a process running on one host to identify a process running on another host? (2 marks)

2. What transport service does an application needed? (2marks)

3. HTTP protocol is stateless. What means the stateless? (mark)

**Question 4:**

1. What is / are the protocol/ protocols you need to transfer and store emails on the receiver's server? (mark)

2. What is/are the protocol/ protocols you need to retrieval emails from server? (mark)

3. What differences between TCP and UDP? (3.5 marks)

| Basis                    | Transmission control protocol (TCP) | User datagram protocol (UDP) |
|--------------------------|-------------------------------------|------------------------------|
| Type of Service          |                                     |                              |
| Reliability              |                                     |                              |
| Error checking mechanism |                                     |                              |
| Acknowledgment           |                                     |                              |
| Speed                    |                                     |                              |
| Retransmission           |                                     |                              |
| Header Length            |                                     |                              |



### Question 5:

(10 marks)

1. User datagram protocol is called connectionless because \_\_\_\_\_  
 a) all UDP packets are treated independently by transport layer      b) it sends data very quickly  
 c) it sends data as a stream of related packets      d) it is received in the same order as sent order
2. Transport layer protocols deals with \_\_\_\_\_  
 a) application to application communication      b) process to process communication  
 c) node to node communication      d) routing
3. In case of rdt2.2 receiver can send \_\_\_\_\_  
 a) Only ACK      b) Only NAK      c) Both ACK and NAK      d) Nothing
4. The checksum is used for \_\_\_\_\_  
 a) data recovery      b) data transfer      c) error detection      d) error recovery
5. A \_\_\_\_\_ protocol provides logical communication between processes running on different hosts.  
 a) transport-layer      b) session-layer      c) network-layer      d) application-layer
6. Negative ACKnowledgements (NAKs) are not needed in a Reliable Data Transfer (RDT) protocol if:  
 a) sequence numbers are being used      b) checksums are being used  
 c) timeouts and retransmissions are being used      d) wireless is being used
7. \_\_\_\_\_ means accepting items from more than one source. \_\_\_\_\_  
 a) Demultiplexing, Multiplexing      b) Multiplexing, Demultiplexing  
 c) Encapsulation, Decapsulation      d) Pulling, Pushing
8. In TCP, the size of the send window is the \_\_\_\_\_ of rwnd and cwnd.  
 a) maximum      b) minimum      c) sum of      d) nothing of them
9. In the GO-Back-N protocol, the maximum send window size is \_\_\_\_\_ and the maximum receive window size is \_\_\_\_\_, where m is related to the number of bits in the sequence number.  
 a) 1, 1      b)  $2^m - 1$ ; 1      c) 1;  $2^m$       d)  $2^m - 1$ ;  $2^m - 1$
10. In the GO-Back-N protocol, the maximum send window size is \_\_\_\_\_ and the maximum receive window size is \_\_\_\_\_, where m is related to the number of bits in the sequence number.  
 a) 1, 1      b) 1;  $2^m$       c)  $2^m - 1$ ;  $2^m - 1$       d)  $2^m - 1$ ; 1

Good luck